System API Document

EPC Lodgement Rules Project
Updated Endpoint Specification

Updated endpoint specification for Scottish EPC Register lodgement

(EPC Lodgement Rules API changes.docx)

Commercial in Confidence

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Document version history

Document Version	Date	Author	Description
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0.2	12th November 2020	Scott Halls	Minor changes to reflect updates – DRAFT
0.3	25th November 2020	Scott Halls	Minor updates – DRAFT
0.4	11th January 2021	Scott Halls	Change to reporting endpoint address - DRAFT
0.5	06th May 2021	Scott Halls	Update to draft EPC lodgement return - DRAFT

1. Introduction

Following a review by Elmhurst Energy Services of historic lodgement data contained within the Scottish EPC Register, a report was produced recommending the development of a new Lodgement Rules sub system within the Scottish EPC Register.

Specifically, this new Lodgement Rules system will add additional data quality checks to EPC lodgements of the following types:

- Domestic RDSAP
- Domestic SAP
- Non Domestic SBEM/DSM

A part of this development requires changing the existing lodgement error response XML, with a view to including multiple errors in the response, as well as detailing any non compliance with the new lodgement rules.

In addition to the new XML response, a new API endpoint is being developed to provide the retrieval of historic lodgement validation data contained within a centrally held validation audit trail.

This document describes the changes in the lodgement XML response, and provides details of the new error audit reporting functionality.

NOTE: This is a DRAFT version of this document. This document can be used to inform the likely changes to the EPC Lodgement API endpoints but should not be treated as final. Any information in this document may change.

2. Changes to lodgement error reporting

2.1 Changes overview

When an EPC lodgement is performed, a number of tests are run to check the quality of the data contained within the lodgement XML.

Current process overview:

1. If the lodgement passes all of the tests, the EPC is lodged and a successful creation status is returned (201 Created).
2. When an error is encountered, the EPC is not lodged and a bad request status is returned (400 Bad request) immediately. All subsequent checks are not tested.

The changes to this existing method are:

- Responses containing single errors have been replaced with responses containing multiple errors.
- The new lodgement rules have been added to the list of errors
- Errors are marked as either:
  - Critical (lodgement was not successful)
  - Warning (lodgement was allowed but with warnings)
- Expanded error information
- A summary record is returned with every response (including successful 201 responses)
- A new Lodgement Instance ID is generated and returned

These changes apply to the Domestic AND Non-Domestic lodgement API endpoints.

The new structure presented here represents a breaking change to the existing legacy format.

These changes are described in more detail below.

2.2 Current lodgement response

The current lodgement process perform a series of tests before lodgement is committed. These tests are carried out in sequence and when an error is encountered a response is immediately returned. As a result, error responses currently only contain a single error.

If a particular lodgement contains multiple errors, then only the first one encountered will be seen in the response. The subsequent errors will only be reported, one at a time, as the proceeding ones are resolved. This could involve multiple lodgement attempts before a successful lodgement is made.

Example of current response, showing singular error.

```
<ErrorResponse xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"xmlns="http://www.est.org.uk/epc/estcrV1/DCLG-HIP">
        <ErrorCode>ER032</ErrorCode>
        <ErrorMessage>Invalid request</ErrorMessage>
</ErrorResponse>
```

These errors are returned with a status code of 400 (Bad request).

Successful lodgements are acknowledged with a 201 (Created) status code, but no response payload is returned.

With the addition of the new lodgement rules, plus the ability to support “warnings”, a new approach is required.

2.3 New response definition

This section provides a description of the new data model used in the validation response.

The new format represents a breaking change to the existing legacy format.

2.3.1 Types of error

All errors will be one of two types:

- **Critical**: Stopped the lodgement of the EPC and returned a status of 400. A Critical error is a fatal test failure that prevents the lodgement.
- **Warning**: Did not stop the lodgement of the EPC. A warning is a failed test that is not significant enough to prevent a lodgement but is reported back to the service consumer.

If there are ANY critical errors reported, then this will prevent the lodgement of the EPC and raise a status of 400, unless they are authentication issues, in which case they will have a status of 403.

2.3.2 Validation response types

The following table describes the various responses to different scenarios.

Scenario	Response code	Response body payload	EPC Lodged?
EPC Lodgement made with no errors	HTTP 201 (Created)	Validation Response XML object containing summary header only	YES
EPC Lodgement made with warnings only	HTTP 201 (Created)	Validation Response XML object containing summary and list of warnings	YES
EPC Lodgement with ANY critical errors	HTTP 400 (Bad request)	Validation Response XML object containing summary and list of critical errors (plus any additional warnings)	NO
EPC Lodgement failed authentication	HTTP 403 (Forbidden)	Validation Response XML object containing summary and reason for authentication failure	NO

Important:
It is important to note that a response body payload will be included for ALL lodgements, including successful ones.

Lodgements that contain non-critical warnings will still be returned with a status code of 201 (Created). Details of any warnings will be contained within the payload. It is the responsibility of the consuming application client to interrogate and show these errors to the user if required.

2.3.3 Object description

All responses will be returned as UTF-16 encoded XML.

All errors will be presented in this format. This includes any current validation errors, as well as those contained within the new lodgement rules.

The expanded Validation Response object is described below:

ValidationResponse object

Field	Data Type	Description
LodgementSuccess	Boolean	Indicates whether the lodgement was successful (contained no Critical Errors)
NumberOfWarnings	Integer	A count of the Non-Critical Warnings in the response
NumberOfCriticalErrors	Integer	A count of the Critical Errors in the response
LodgementInstanceID	String GUID ID	A unique reference to this lodgement attempt. Please see below.
ErrorList	List of type **Error**	List containing warnings and errors as defined below.

Error object

Field	Data Type	Description
ErrorSeverity	String	Contains either: • "Critical", or • "Warning"
ErrorID	String	Error ID reference number
ErrorField	String	A friendly description of the field in the lodgement that caused the error (if applicable)
ErrorMessage	String	Details of the error
ErrorFieldPath	String	Explicit XML path to the field in the lodgement that caused the error (if applicable)
SuppliedValue	String	The value in the field that cause the error (if applicable)
Comment	String	Supplementary comments returned by the validation logic.

2.3.4 Batch errors versus Singular errors

Whilst the majority of the potential errors can be returned as a batch, there are a few errors that have to be returned individually. These represent fatal errors that impede further steps and validation.

These checks are carried out first, and if any of them fail a response is immediately returned with a status of 400, or 403. Details of the error will be contained in the Validation Response object as detailed above.

These singular errors contain, but are not limited to:

- The format of the EPC payload is incorrect in some fundamental way making further interrogation invalid;
- Authentication errors;
- Incorrect EPC Schema version;
- RRN errors and mismatches;

2.3.5 Field paths

If an error relates to a specific field within the EPC XML a formatted path will be returned within the **ErrorFieldPath** field of the **Error** object where possible.

This contains an explicit path to the location of the field that failed the test.

Example paths:

```
/RdSAP-Report[1]/SAP-Data[1]/SAP-Property-Details[1]/Multiple-Glazed-Proportion
```

```
/RdSAP-Report[1]/SAP-Data[1]/SAP-Property-Details[1]/SAP-Building-Parts[1]/SAP-Building-Part[2]/Floor-U-Value
```

The numbers contained within square brackets (**[n]**) represent the indexed instance of that node within the parent node of the XML structure.

In the first example above, all nodes are the first child node of that type within the parent node.

In the second example SAP-Building-Part[2] is the second child of the parent node SAP-Building-Parts[1].

This is especially useful to identify errors in nodes that may have other instances within the structure (such as SAP-Building-Part or SAP-Floor-Dimension).

This is not possible for all errors.

2.3.6 Lodgement Instance ID

All responses will now include a **LodgementInstanceID**. This represents the specific attempt at lodgement within the Scottish EPC Register. Each new attempt at lodgement, irrespective of the property, will be given a new instance ID.

The Instance ID can be used with the new audit endpoint as outlined below in section 3.

2.3.7 Example XML output

Please see below two XML examples of the new format

Example 1: Output with 2 critical errors

```
<ValidationResponse xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://www.est.org.uk/epc/estcrV2/DCLG-HIP">

  <LodgementSuccess>false</LodgementSuccess>
  <NumberOfWarnings>0</NumberOfWarnings>
  <NumberOfCriticalErrors>2</NumberOfCriticalErrors>
  <LodgementInstanceID>fc05bc6e-6cfd-4753-96e2-22e5b8d69555</LodgementInstanceID>

  <ErrorList>
    <ErrorItem>
      <ErrorSeverity>Critical</ErrorSeverity>
      <ErrorID>ERXXX</ErrorID>
      <ErrorField>Property address - Postcode</ErrorField>
      <ErrorMessage>Value cannot be null</ErrorMessage>
      <ErrorFieldPath>/RdSAP-Report[1]/Report-
Header[1]/Property[1]/Address[1]/Postcode</ErrorFieldPath>
      <SuppliedValue>NULL</SuppliedValue>
      <Comment>Example comment</Comment>
    </ErrorItem>
    <ErrorItem>
      <ErrorSeverity>Critical</ErrorSeverity>
      <ErrorID>ERXXX</ErrorID>
      <ErrorField>SAP Property Details - Multiple-Glazed-Proportion</ErrorField>
      <ErrorMessage>Value must be less than or equal to 100</ErrorMessage>
      <ErrorFieldPath>/RdSAP-Report[1]/SAP-Data[1]/SAP-Property-Details[1]/Multiple-
Glazed-Proportion</ErrorFieldPath>
      <SuppliedValue>120</SuppliedValue>
```

```
    </ErrorItem>
  </ErrorList>

</ValidationResponse>
```

Example 2: Output with no errors and no warnings

```
<?xml version="1.0" encoding="utf-16"?>
<ValidationResponse xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://www.est.org.uk/epc/estcrV2/DCLG-HIP">

  <LodgementSuccess>true</LodgementSuccess>
  <NumberOfWarnings>0</NumberOfWarnings>
  <NumberOfCriticalErrors>0</NumberOfCriticalErrors>
  <LodgementInstanceID>68090e78-1b94-41a2-9e47-ecf1c110c595</LodgementInstanceID>

  <ErrorList />

</ValidationResponse>
```

2.3.8 Changes to Draft EPC lodgement

2.3.8.1 Current mechanism

In the current version of the register a successful submission of a valid EPC to the draft domestic endpoint (as listed in section 2.4) results in a draft EPC PDF being returned within the response body encoded as a Base 64 string. This can then be rendered/saved as required by the client application. However, if an error is found, this is returned in the response body with the appropriate HTTP status code.

2.3.8.2 New mechanism

The new endpoint can return both critical errors and/or warnings. If critical errors are present, the structure described above in section 2.3 is adequate as no draft EPC PDF is returned.

However, is the EPC only contains warnings it is necessary to return both the warnings and the draft EPC within the same response.

In this case the ValidationResponse object described in section 2.3 is expanded to contain an additional field entitled “PDFContent” which will contain a Base 64 encoded string of the draft EPC PDF. This field will only be populated if the draft submission did not contain any errors marked as Critical.

Extension of ValidationResponse definition

Field	Data Type	Description
PDFContent	String	Base 64 encoded string containing draft EPC report.

2.4 New endpoints and legacy support

The new validation methodology will be implemented for the following types of lodgement:

- Domestic EPC lodgement (RdSAP and SAP)
- Draft Domestic EPC lodgement (RdSAP and SAP)
- Non-Domestic EPC lodgement (SBEM)

To facilitate a smooth transition to the new mechanism, a set of new endpoints will be made available.

The new endpoints will be:

```
LIVE
https://dom.scottishepcregister.org.uk/EPCV2/RRN/{id}
https://dom.scottishepcregister.org.uk/DraftEPCV2/RRN/{id}
https://nondom.scottishepcregister.org.uk/EPCV2/RRN/{id}
https://dom.scottishepcregister.org.uk/EPCNotForIssueV2/RRN/{RRN}
https://nondom.scottishepcregister.org.uk/EPCNotForIssueV2/RRN/{RRN}

UAT
https://domuat1.solsticecloud.com/EPCV2/RRN/{id}
https://domuat1.solsticecloud.com/DraftEPCV2/RRN/{id}
https://nondomuat1.solsticecloud.com/EPCV2/RRN/{id}
https://domuat1.solsticecloud.com /EPCNotForIssueV2/RRN/{RRN}
https://nondomuat1.solsticecloud.com/EPCNotForIssueV2/RRN/{RRN}
```

The version 1 endpoints will initially remain unchanged.

All other existing functionality at the legacy locations (`/DraftEPC/` & `/EPC/`) will be brought across to the new endpoints. For example `/EPC/Postcode/{postcode}` will be made available on `/EPCV2/Postcode/{postcode}`. Error responses to these existing functions will remain in the existing format. With the exception of the new lodgements process, all other functionality will remain unchanged.

Whilst transition to the new format is ongoing both sets of endpoints will accept requests. During transition, all changes to the legacy endpoints will be locked.

After transition to the new format is complete, and all client applications have been updated to accept this new format, the legacy endpoints will be retired.

UPODATE: PLEASE NOTE THE VERSION 1 ENDPOINTS ARE NOW UNAVAILABLE

3 New Audit Report API

3.1 Purpose of the API

An audit trail of all lodgement errors will be maintained within the central Scottish EPC register. This will record all critical errors and warnings generated during a lodgement.

A new API endpoint will be provided to allow approved organisations to download a list of errors stored within the audit trail.

Specifically the following functions will be provided:

- Get Errors by assessor ID and Date Range
- Get Errors by UPRN
- Get Errors by RRN
- Get Errors by Lodgement Attempt ID

3.2 Endpoint location

The root API will be made available at :

```
LIVE
https://dom.scottishepcregister.org.uk/EPCV2/AuditReport/...
and
https://nondom.scottishepcregister.org.uk/EPCV2/AuditReport/..

UAT
https://domuat1.solsticecloud.com/EPCV2/AuditReport/...
and
https://nondomuat1.solsticecloud.com/EPCV2/AuditReport/...
```

Calls to this root address will not yield any results. Please see section 3.4 for a list of specific calls.

3.3 Standard response object

All requests to this API will return data in a standard format as outlined below.

AuditReportHeader object

Field	Data Type	Description
Authenticated	Boolean	Was the call authenticated
Error	String	Details of any error
NumberOfLodgements	Int	Integer indicating the number of lodgements attempts included in the response.
LodgementAuditList	List of type LodgementAuditInstance	A list of lodgement attempts that match the criteria specified

LodgementAuditInstance object

Field	Data Type	Description
LodgementInstanceID	String GUID ID	A unique reference to this lodgement attempt.
MethodologyType	String	RdSAP, SAP, SBEM

IsDraftEPC	Boolean	Flag indicating whether the lodgement was to the DRAFT endpoint.
LodgementSuccess	Boolean	Indicates whether the lodgement was successful (contained no Critical Errors)
NumberOfWarnings	Integer	A count of the Non-Critical Warnings in the response
NumberOfCriticalErrors	Integer	A count of the Critical Errors in the response
AssessorID	String	ID of the lodging assessor
AssessorName	String	Name of the lodging assessor
UPRN	String	The UPRN associated with the lodgement
RRN	String	The RRN associated with the lodgement
LogDate	Datetime	Date and time stamp of the lodgement attempt
ErrorList	List of type **Error**	List containing warnings and errors as in section 2.3

Example XML response


```xml
<?xml version="1.0" encoding="utf-16"?>
<AuditReportHeader xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://www.est.org.uk/epc/estcrV2/DCLG-HIP">

  <Authenticated>True</Authenticated>
  <Error></Error>
  <NumberOfLodgements>2</NumberOfLodgements>
  <LodgementAuditList>
    <LodgementAuditInstance>
      <LodgementInstanceID>fc05bc6e-6cfd-4753-96e2-22e5b8d69555</LodgementInstanceID>
      <MethodologyType>SAP</MethodologyType>
      <IsDraftEPC>false</IsDraftEPC>
      <LodgementSuccess>false</LodgementSuccess>
      <NumberOfWarnings>0</NumberOfWarnings>
      <NumberOfCriticalErrors>2</NumberOfCriticalErrors>
      <AssessorID>XXXXX</AssessorID>
      <AssessorName>NNNNNNNN</AssessorName>
      <UPRN>xxxxxxxx<</UPRN>
      <RRN>xxxx-xxxx-xxxx-xxxx-xxxx</RRN>
      <LogDate>dd-MMM-yyyy hh:mm:ss</LogDate>

      <ErrorList>
        <Error>
          <ErrorSeverity>Critical</ErrorSeverity>
          <ErrorID>ERXXXX</ErrorID>
          <ErrorField>Property address - Postcode</ErrorField>
          <ErrorMessage>Value cannot be null</ErrorMessage>
          <ErrorFieldPath>/RdSAP-Report[1]/Report-
Header[1]/Property[1]/Address[1]/Postcode</ErrorFieldPath>
          <SuppliedValue>NULL</SuppliedValue>
        </Error>
        <Error>
          <ErrorSeverity>Critical</ErrorSeverity>
          <ErrorID>ERXXXX</ErrorID>
          <ErrorField>SAP Property Details - Multiple-Glazed-Proportion</ErrorField>
          <ErrorMessage>Value must be less than or equal to 100</ErrorMessage>
          <ErrorFieldPath>/RdSAP-Report[1]/SAP-Data[1]/SAP-Property-
Details[1]/Multiple-Glazed-Proportion</ErrorFieldPath>
          <SuppliedValue>120</SuppliedValue>
        </Error>
      </ErrorList>
    </LodgementAuditInstance>

    <LodgementAuditInstance>
      ...
    </LodgementAuditInstance>

  </LodgementAuditList>
```


The individual function calls are detailed below.

3.4 Individual function calls

3.4.1 Get audit by Assessor Membership Number and date range

Returns a list of audits, and associated errors, for a given Assessor and date range.

Request

URL	**LIVE** GET https://dom.scottishepcregister.org.uk/EPCV2/AuditReport/Assessor/ HTTP/1.1 GET https://nondom.scottishepcregister.org.uk/EPCV2/AuditReport/Assessor/ HTTP/1.1 **UAT** GET https://domuat1.solsticecloud.com/EPCV2/AuditReport/Assessor/ HTTP/1.1 GET https://nondomuat1.solsticecloud.com/EPCV2/AuditReport/Assessor/ HTTP/1.1
Header	Authorization: Basic <Base64 UTF-16 username:password> Assessor: <Base64 UTF-16 Assessor Membership Number> FromDate: <yyyy-MM-dd> ToDate: <yyyy-MM-dd>

NOTE : {membershipnumber} needs to be URL encoded

Logic

Audit information will be returned where:

- It matches the requested Assessor Membership Number
- It was after or equal to the FromDate
- It was before or equal to the ToDate
- The user details specified in Authorization are valid
- The requested lodgement had errors or warnings. Successful lodgements with no errors and warnings will not be returned

Response

Result	Response Type
Successfully retrieved result	UTF-16 encoded XML of type **AuditReportHeader** outlined in 3.3
Error	UTF-16 encoded XML of type **AuditReportHeader** outlined in 3.3 detailing error

3.4.2 Get audit by UPRN

Returns a list of audits, and associated errors, for a given UPRN.

Request

URL	**LIVE** GET https://dom.scottishepcregister.org.uk/EPCV2/AuditReport/UPRN/{**UPRN**} HTTP/1.1 GET https://nondom.scottishepcregister.org.uk/EPCV2/AuditReport/UPRN/{**UPRN**} HTTP/1.1 **UAT** GET https://domuat1.solsticecloud.com/EPCV2/AuditReport/UPRN/{**UPRN**} HTTP/1.1 GET https://nondomuat1.solsticecloud.com/EPCV2/AuditReport/UPRN/{**UPRN**} HTTP/1.1
Header	Authorization: Basic <Base64 UTF-16 username:password>

Logic

Audit information will be returned where:

- It matches the requested UPRN
- The user details specified in the Authorization are valid
- The requested lodgement had errors or warnings. Successful lodgements with no errors and warnings will not be returned

Response

Result	Response Type
Successfully retrieved result	UTF-16 encoded XML of type **AuditReportHeader** outlined in 3.3
Error	UTF-16 encoded XML of type **AuditReportHeader** outlined in 3.3 detailing error

3.4.3 Get audit by RRN

Returns a list of audits, and associated errors, for a given RRN.

Request

URL	**LIVE** GET https://dom.scottishepcregister.org.uk/EPCV2/AuditReport/RRN/**{RRN}** HTTP/1.1 GET https://nondom.scottishepcregister.org.uk/EPCV2/AuditReport/RRN/**{RRN}** HTTP/1.1 **UAT** GET https://domuat1.solsticecloud.com/EPCV2/AuditReport/RRN/**{RRN}** HTTP/1.1 GET https://nondomuat1.solsticecloud.com/EPCV2/AuditReport/RRN/**{RRN}** HTTP/1.1
Header	Authorization: Basic <Base64 UTF-16 username:password>

Logic

Audit information will be returned where:

- It matches the requested RRN
- The user details specified in the Authorization are valid
- The requested lodgement had errors or warnings. Successful lodgements with no errors and warnings will not be returned

Response

Result	Response Type
Successfully retrieved result	UTF-16 encoded XML of type **AuditReportHeader** outlined in 3.3
Error	UTF-16 encoded XML of type **AuditReportHeader** outlined in 3.3 detailing error

3.4.4 Get audit by Lodgement Instance ID

Returns details of a specific instance of an lodgement audit, along with any associated errors.

Request

URL	**LIVE** GET https://dom.scottishepcregister.org.uk/EPCV2/AuditReport/InstanceID/{**InstanceID**} HTTP/1.1 GET https://nondom.scottishepcregister.org.uk/EPCV2/AuditReport/InstanceID/{**InstanceID**} HTTP/1.1 **UAT** GET https://domuat1.solsticecloud.com/EPCV2/AuditReport/InstanceID/{**InstanceID**} HTTP/1.1 GET https://nondomuat1.solsticecloud.com/EPCV2/AuditReport/InstanceID/{**InstanceID**} HTTP/1.1
Header	Authorization: Basic <Base64 UTF-16 username:password>

Logic

Audit information will be returned where:

- It matches the requested InstanceID
- The user details specified in the Authorization are valid
- The requested lodgement had errors or warnings. Successful lodgements with no errors and warnings will not be returned.

Response

Result	Response Type
Successfully retrieved result	UTF-16 encoded XML of type **AuditReportHeader** outlined in 3.3
Error	UTF-16 encoded XML of type **AuditReportHeader** outlined in 3.3 detailing error

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